

AI in Disaster Management

When Machines Become Lifesavers

Harshitha D Gowda

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"Predict. Prepare. Protect."

Hook: Imagine if disasters had a spoiler alert... and AI leaked it early!

Problem Statement: The Chaos Problem

Story: Disasters are like surprise exams—no warning, high pressure, and everyone panics.

Challenges:

- Natural disasters cause massive damage
- Traditional systems are slow and reactive
- Data is available but insights are limited

Research Gap:

- Need for early prediction
- Efficient resource allocation
- Real-time decision making

Analogy: AI is like a super-smart weather auntie predicting everything!

- ➊ Data Collection (satellites, sensors, social media)
- ➋ Data Preprocessing (cleaning messy data)
- ➌ Feature Extraction
- ➍ Model Training (ML/DL)
- ➎ Prediction and Alerts

Approach:

- Supervised Learning
- Real-time Analytics

Narrative: AI without data is like a chef without ingredients.

Data Sources:

- Satellite Imagery
- Sensor Data (temperature, seismic)
- Historical Records
- Social Media Feeds

Challenges:

- Missing Data
- Noise in social media
- Data inconsistency

Analogy: AI models are detectives solving mysteries before they happen.

Models Used:

- Machine Learning: Decision Trees, Random Forest
- Deep Learning: CNN, RNN, LSTM

Workflow:

Input Data → Model → Prediction → Alert

Insight: AI isn't just smart—it's scary accurate!

Outcomes:

- Prediction accuracy improved (85–95%)
- Faster response time
- Better resource allocation

Impact:

- Early warnings save lives
- Reduced economic losses

Narrative: AI is that friend who says “I told you so” — but actually helps!

Key Takeaways:

- AI enables proactive disaster management
- Improves decision-making
- Supports real-time insights

Future Scope:

- Integration with IoT
- Autonomous rescue systems
- Improved real-time prediction models

Questions?

Fun Line: No question is a disaster—unless you don't ask it!

Discussion Prompts:

- How helpful is AI in detecting disaster ?
- Can AI replace human decision-making?